

RMO– 2016(2)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABC be a triangle and D be the mid point of BC. Suppose the angle bisector of $\angle ADC$ is tangent to the circumcircle of triangle ABD at D. Prove that $\angle A = 90^\circ$.
 2. Let a, b, c be three distinct positive real numbers such that $abc = 1$. Prove that:

$$\frac{a^3}{(a-b)(a-c)} + \frac{b^3}{(b-c)(b-a)} + \frac{c^3}{(c-a)(c-b)} \geq 3.$$
 3. Let a, b, c, d, e, f be positive integers such that $\frac{a}{b} < \frac{c}{d} < \frac{e}{f}$. Suppose that $af - be = -1$, show that $d \geq b + f$.
 4. There are 100 countries participating in an Olympiad. Suppose n is a positive integer such that each of the 100 countries is willing to communicate in exactly n languages. If each set of 20 countries can communicate in at least one common language, and no language is common to all 100 countries, what is the minimum possible values of n ?
 5. Let ABC be a right angled triangle with $\angle B = 90^\circ$. Let I be the incentre of ABC. Extend AI and CI; let them intersect BC in D and AB in E respectively. Draw a line perpendicular to AI at I to meet AC in J; draw a line perpendicular to CI at I to meet AC in K. Suppose $DJ = EK$. Prove that $BA = BC$.
 6.
 - (a) Given any natural number N, prove that there exists a strictly increasing sequence of N positive integers in harmonic progression.
 - (b) Prove that there cannot exist a strictly increasing infinite sequence of positive integers which is in harmonic progression.

**